Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

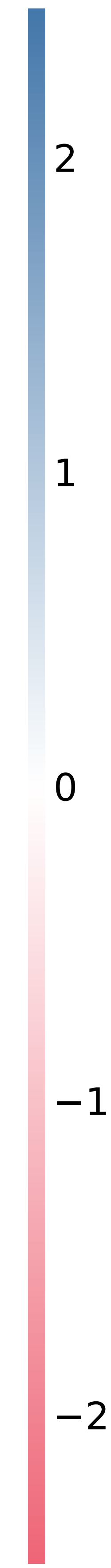
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV



Reasoning + Answer:

<think>

de Broglie wavelength of a particle is calculated using the formula Î» = h/p, where Î» is the wavelength and h is Planck's constant. Given the kinetic energy of the electron (E_k = 1eV), we need to calculate its momentum p first.

To calculate the momentum, we'll need to convert the kinetic energy from eV to Joules, as Planck's constant (h) is in units of Js.

1eV = 1.602 ÃĹ 10^-19 J

1eV kinetic energy is (1.602 ÃĹ 10^-19 J = 1.602 ÃĹ 10^-19 J

so for 1eV energy 1eV kinetic energy = 1.602 ÃĹ 10^-19 J

|p| = sqrt(2m ÃĹ 1eV kinetic energy) = sqrt(2 ÃĹ 1.602 ÃĹ 10^-19 J) = 4.899 ÃĹ 10^-10 kgm/s

Î» = h/p

Planck constant h = 6.626 x 10^-34 Jmn

Î» = (6.626 * 10^-34 Jm m)/4.899 x 10^-10 dm

Î» = 1.34 ÃĹ 10^-34 = 1.34 âĪĵ 10 measure